

Product Laws of Exponents

MA.7.NSO.1.1

I can use the product of powers, power of a product, power of a power and identity laws to rewrite and evaluate a numerical expression.

Last unit you took roots apart.

Today you put powers together — and the base has to match.

Warm-Up

On your own. Four minutes. No notes.

1. $\sqrt{81}$

2. Between which two whole numbers is $\sqrt{40}$?

3. Solve $3x - 7 = 11$

4. Evaluate $\left(\frac{2}{3}\right)^2$

Warm-Up

Answers.

1. $\sqrt{81}$ **9**

2. Between which two whole numbers is $\sqrt{40}$? **6 and 7**

3. Solve $3x - 7 = 11$ **$x = 6$**

4. Evaluate $\left(\frac{2}{3}\right)^2$ **$\frac{4}{9}$**

Notes

Copy the three words.

I. A power is a shorthand for repeated multiplication

A. **base** — the number used as a factor

B. **exponent** — how many times the base is a factor

C. **power** — the whole thing, base and exponent together

$$2^3 = 2 \cdot 2 \cdot 2 = 8 \qquad \left(\frac{2}{3}\right)^2 = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$$

Notes

Where the rule comes from.

II. Product of powers — same base, add the exponents

$$3^2 \cdot 3^3 = (3 \cdot 3)(3 \cdot 3 \cdot 3) = 3^5$$

$$3^2 \cdot 2^3 = (3 \cdot 3)(2 \cdot 2 \cdot 2) \quad \text{— different bases, no shortcut}$$

Product of powers: $a^m \cdot a^n = a^{m+n}$

Check with values: $3^2 \cdot 3^3 = 9 \cdot 27 = 243 = 3^5$

Notes

III. Power of a product — the exponent goes to every factor

$$(2 \cdot 5)^3 = (2 \cdot 5)(2 \cdot 5)(2 \cdot 5) = (2 \cdot 2 \cdot 2)(5 \cdot 5 \cdot 5) = 2^3 \cdot 5^3$$

Power of a product: $(ab)^m = a^m \cdot b^m$

Works for a product only, never for a sum:

$$(10 + (-2))^3 = 8^3 = 512 \quad \text{but} \quad 10^3 + (-2)^3 = 992$$

Notes

IV. Power of a power, and the exponent 1

Power of a power	Repeated multiplication	Single power
$(4^2)^3$	$4^2 \cdot 4^2 \cdot 4^2$	4^6
$(10^4)^4$	$10^4 \cdot 10^4 \cdot 10^4 \cdot 10^4$	10^{16}
$(3^1)^5$	$3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$	3^5

Power of a power: $(a^m)^n = a^{mn}$ **Identity:** $a^1 = a$

Example 1

Two ways to the same value.

$$3^2 \cdot 3^3$$

Which law? Then find the value.

Example 1

Worked both ways.

$$3^2 \cdot 3^3 = 3^{2+3} = 3^5 \quad \text{product of powers: same base, add}$$

$$3^5 = 243 \quad \text{evaluate the single power}$$

$$9 \cdot 27 = 243 \quad \text{check: evaluate each, then multiply}$$

Answer: 243

Your Turn

Same steps, your numbers. Boards up when done.

$$2^3 \cdot 2^4$$

Your Turn

Answer.

$$2^3 \cdot 2^4$$

same base, add: 2^7

$$\textbf{Answer: } 2^7 = 128$$

Example 2

A negative base and a power of a power.

$$[(-2)^3]^2$$

Which law? Then find the value.

Example 2

Worked.

$$[(-2)^3]^2 = (-2)^{3 \cdot 2} = (-2)^6 \quad \text{power of a power: multiply}$$

$$(-2)^6 = 64 \quad \text{six negative factors — positive}$$

$$(-8)^2 = 64 \quad \text{check: } (-2)^3 = -8, \text{ then square}$$

Answer: 64

Your Turn

Same steps, your numbers. Boards up when done.

$$[(-3)^2]^3$$

Your Turn

Answer.

$$[(-3)^2]^3$$

multiply the exponents: $(-3)^6$

Answer: $(-3)^6 = 729$

Whiteboards • Question 1 of 9

Boards up on three.

$$5^2 \cdot 5^3$$

Answer it.

Write it as one power.

Whiteboards • Question 1 of 9

Answer.

$$5^2 \cdot 5^3$$

same base, so add the exponents

Answer: 5^5

Whiteboards • Question 2 of 9

Boards up on three.

A photo is enlarged 2^4 times, then 2^2 times more.

In all it is $2^4 \cdot 2^2$ times bigger. The photo is 9 cm wide.

Write $2^4 \cdot 2^2$ as one power. Then find its value.

Answer it.

Whiteboards • Question 2 of 9

Answer.

A photo is enlarged 2^4 times, then 2^2 times more.

In all it is $2^4 \cdot 2^2$ times bigger. The photo is 9 cm wide.

Write $2^4 \cdot 2^2$ as one power. Then find its value.

$$2^4 \cdot 2^2 = 2^6$$

Answer: $2^6 = 64$

Whiteboards • Question 3 of 9

Boards up on three.

$$(2 \cdot 5)^3$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$(2 \cdot 5)^3$$

$$2^3 \cdot 5^3 = 8 \cdot 125, \text{ or } 10^3$$

Answer: 1000

Whiteboards • Question 4 of 9

Boards up on three.

$$(3^2)^4$$

Answer it.

Write it as one power.

Whiteboards • Question 4 of 9

Answer.

$$(3^2)^4$$

four copies of 3^2 , so eight 3s

Answer: 3^8

Whiteboards • Question 5 of 9

Boards up on three.

$$4^3 \cdot 4^2$$

Which expression is equivalent?

A. 16^5

B. 8^5

C. 4^6

D. 4^5

Answer it.

Whiteboards · Question 5 of 9

Answer.

$$4^3 \cdot 4^2$$

Which expression is equivalent?

A. 16^5

B. 8^5

C. 4^6

D. 4^5

Answer: D — 4^5

Whiteboards • Question 6 of 9

Boards up on three.

$$(2^3)^2$$

Which expression is equivalent?

A. 2^5

B. 4^6

C. 2^6

D. 2^9

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$(2^3)^2$$

Which expression is equivalent?

A. 2^5

B. 4^6

C. 2^6

D. 2^9

Answer: C — 2^6

Whiteboards • Question 7 of 9

Boards up on three.

$$(-3)^2 \cdot (-3)^1$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$(-3)^2 \cdot (-3)^1$$

$(-3)^3$ — three negative factors

Answer: -27

Whiteboards • Question 8 of 9

Boards up on three.

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$$

Answer it.

Find the value.

Whiteboards • Question 8 of 9

Answer.

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$$
$$\left(\frac{1}{2}\right)^5$$

Answer: $\frac{1}{32}$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Aiyanna says she can use the power of a product law to evaluate $(10 + (-2))^3$. Is she right?

Write your answer in sentences.

Say what the law needs, and show the value.

Whiteboards • Question 9 of 9

Answer.

Aiyanna says she can use the power of a product law to evaluate $(10 + (-2))^3$. Is she right?

the parentheses hold a SUM, not a product

Answer: No — $(10 + (-2))^3 = 8^3 = 512$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate powers — SZQ

Multiplication with exponents — 5SK

Power rule — 8DA

Understanding exponents — BFA